ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 2 · PRACTICE D

MARKS: 80 **TIME: 4 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET second paper (Raspberry Pi practical, 80 marks, 4 hours): a reaction-time game with breadboard and flowchart, completing and commenting a door-alarm program, and an automatic street light with breadboard, Python code and a user manual.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of THREE questions. Answer ALL the questions.
2. These questions are done on a computer, using draw.io (or similar) for flowcharts, Fritzing for breadboard circuits, a Python 3 IDE for code and a word processor for documents. If you do not have a computer, draw and write your answers neatly by hand.
3. Use a Raspberry Pi (any model with a 40-pin header). Use the BCM (GPIO) numbers of the pins.
4. Put your name or examination number on every printout or page. Save your work regularly.

QUESTION 1

> A science club wants a reaction-time game for its open day. A yellow LED switches on after a random wait of 2 to 5 seconds. The player must press a button as quickly as possible when the LED comes on. The program shows the reaction time, plays three rounds and then shows the best (fastest) time. If the player presses the button before the LED comes on, the round counts as "Too early!".

Components:

- Raspberry Pi
- 1 × yellow LED and 1 × 220 Ω resistor
- 1 × push button
- Breadboard and jumper wires

1.1 Using the components above, create a breadboard circuit for the reaction-time game (use Fritzing). Connect the button between a GPIO pin and GND (the internal pull-up will be used). (10)

1.2 Create a flowchart of the logic of the game, including the random wait, the "too early" check, measuring the reaction time, the three rounds and the best time. (11)

1.3 Save and print (or neatly present) the breadboard circuit and the flowchart. (2)

[23]

QUESTION 2

> A small shop wants a door alarm. A magnetic reed switch on the door is connected to GPIO 5 and reads LOW while the door is closed. An ARM button on GPIO 6 switches the alarm on. When the alarm is armed and the door opens, a red LED on GPIO 13 and a buzzer on GPIO 19 must switch on. Pressing the ARM button again switches the alarm off.

Below is the program with ELEVEN parts missing, marked …(1)… to …(11)…. Analyse the code and answer the questions.

```
import RPi.GPIO as GPIO
import time

DOOR = 5
```

```
ARM = 6
LED = 13
BUZZER = 19

GPIO.setmode(GPIO.BCM)
GPIO.setup(DOOR, GPIO.IN, pull_up_down=GPIO.…(1)…)
GPIO.setup(ARM, GPIO.IN, pull_up_down=GPIO.PUD_UP)
GPIO.setup([LED, BUZZER], GPIO.…(2)…)

armed = …(3)…
try:
    while True:
        if GPIO.input(ARM) == GPIO.…(4)…:
            armed = not …(5)…
            print("Armed" if armed else "Disarmed")
            time.sleep(…(6)…)
        if armed and GPIO.input(DOOR) == GPIO.…(7)…:
            GPIO.output(LED, GPIO.HIGH)
            GPIO.output(BUZZER, GPIO.…(8)…)
        else:
            GPIO.output(LED, GPIO.…(9)…)
            GPIO.output(BUZZER, GPIO.LOW)
        time.sleep(0.05)
except …(10)…:
    print("Alarm program stopped")
finally:
    GPIO.…(11)…()
```

2.1 Open a Python IDE and create a new file called Question2_YourName.py. (2)

2.2 Recreate the program, filling in all the missing parts so that it works. (11)

2.3 Add comments to the code that explain what it does, including the pin set-up, arming and disarming, the alarm decision and the exception handling. (10)

2.4 Save and print (or neatly present) the code. (2)

[25]

QUESTION 3

A village wants street lights that switch on by themselves at night. An LDR light-sensor module gives a digital output (DO) that goes HIGH when it is dark. When it is dark, two white LEDs (the street lights) must switch on. A green LED shows that the system has power and is working. To save power, the street lights must stay off for at least 10 seconds after they switch off, so that they do not flicker when car headlights shine on the sensor.

You have the following components:

- 1 × Raspberry Pi
- 1 × LDR light-sensor module (VCC, GND, DO)
- 2 × white LEDs and 1 × green LED
- 3 × 220 Ω resistors
- Breadboard and jumper wires

3.1 Using the components above, create a breadboard circuit for the automatic street light (use Fritzing). (10)

3.2 Open a Python IDE and create a new file called Question3_YourName.py. (2)

3.3 Write the Python code for the automatic street light. (10)

3.4 Write a short user manual for the street light. It must include: the name and a description of the system; basic operation; troubleshooting; and frequently asked questions (FAQs). (10)

[32]

TOTAL: 80

END OF QUESTION PAPER